Answer Key — Competitive Exam Practice 2: Computer-System Organization and Architecture

GATE (Computer Science) Pattern

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Concepts That Click

Q1 (C). A system call is a deliberate, synchronous request and enters through a trap; a disk completion is asynchronous and enters through an interrupt.

(A) False — the two are reversed: interrupts are asynchronous and traps synchronous.

(B) False — divide-by-zero is a synchronous exception (trap) caused by the instruction, not a hardware interrupt.

(C) True — trap for the system call, interrupt for the device signal.

(D) False — only traps are tied to a specific instruction; an interrupt can arrive between any two instructions, at a point the process does not choose.

Q2 4416. Entry i begins at $4096+8i$, so for $i = 40$ the address is $4096+8\times40 = 4096+320 = 4416$.

Q3 5. Cost per interrupt $= 4.5+8 = 12.5\ \mu$s. In one second, $4{,}000\times12.5\ \mu\text{s} = 50{,}000\ \mu\text{s} = 0.05$ s. Fraction of CPU time $= 0.05/1 = 0.05 = 5\%$.

Q4 (C). A lower-priority request remains pending while it is masked and is normally serviced once the higher-priority handler completes and unmasks it.

(A) False — the interrupt-enable flag masks maskable interrupts only; non-maskable interrupts bypass it.

(B) False — masking defers delivery; the request stays pending (level-triggered) or latched (edge-triggered) and can be serviced later.

(C) True — deferral, then normal service at the request's own priority level.

(D) False — priorities are assigned by the interrupt-controller hardware (and can be programmed), not by software installation order.

Q5 10.20. Rotation period $= 60/7{,}200$ s $= 8.333$ ms, so average rotational latency $= 8.333/2 = 4.167$ ms. Transfer time $= 4096$ B$/128\times10^6$ B/s $= 32\ \mu$s $= 0.032$ ms. Total $= 6 + 4.167 + 0.032 = 10.199 \approx 10.20$ ms.

Q6 (C). Registers and DRAM are volatile; magnetic disk and NAND flash are non-volatile and retain their contents without power.

(A) False — DRAM is volatile, so variables in memory do not survive a power failure.

(B) False — SRAM is also volatile; both SRAM and DRAM lose their contents without power.

(C) True — the correct split between volatile and non-volatile levels.

(D) False — NAND flash is non-volatile (its cost-per-byte claim is irrelevant to the error).

Q7 **3.** AMAT = L1 hit time + (L1 miss rate) × [L2 hit time + (L2 miss rate) × memory time]. L1 miss rate is 0.10; L2 miss rate is 0.20. So AMAT = $1 + 0.10 \times (8 + 0.20 \times 60) = 1 + 0.10 \times 20 = 1 + 2 = 3$ ns. The high hit rates are the payoff of temporal and spatial locality.

Q8 **(B).** The device raises an interrupt when it needs attention, so the CPU pays for events rather than spinning on a status register.

(A) False — polling repeatedly reads the status register; it can detect readiness, it merely wastes cycles while waiting.

(B) True — event-driven notification, with CPU cost per event.

(C) False — DMA writes memory behind the CPU's caches, so stale cache lines must be invalidated or flushed; the OS and hardware must maintain coherence.

(D) False — DMA setup is far costlier than one interrupt, so it wins on large block transfers, not on a single byte.

Q9 **0.03.** CPU cost = 2,000 + 1,000 = 3,000 cycles = $3{,}000/(2 \times 10^9)$ s = 1.5 μs. Transfer time = 5 ms = 5,000 μs. Percentage = $(1.5/5{,}000) \times 100 = 0.03\%$.

Q10 **(C).** A cluster is a set of complete, networked machines that share storage and can fail over from one machine to another.

(A) False — each core has its own architectural registers; cores share caches and memory, not registers.

(B) False — SMP is defined by processors sharing one main memory and one OS image.

(C) True — network-connected independent machines, with whole-machine failure tolerance.

(D) False — that describes a multicore chip, not a cluster.

Concepts That Click